DEMO / SYNTHETIC: Effect of laser power and scanning speed on porosity and tensile properties of LPBF Ti-6Al-4V

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Abstract

DEMO / SYNTHETIC TEXT generated to test the AM-KG pipeline; the numbers are not real research results. Ti-6Al-4V specimens were manufactured using laser powder bed fusion (LPBF) with a laser power of 200 W, scanning speed of 800 mm/s and layer thickness of 30 µm. The resulting tensile strength was 950 MPa. Increasing laser power reduced porosity, while a higher scanning speed increased lack of fusion. Hot isostatic pressing (HIP) improved the elongation of the specimens.

1. Introduction

Selective laser melting (SLM), also called laser powder bed fusion, is widely used to process titanium alloys such as Ti6Al4V. Earlier studies reported that the volumetric energy density influences porosity in SLM parts.

2. Materials and methods

2.1 Powder and LPBF process

Gas-atomised Ti–6Al–4V powder with a powder particle size of 15–45 µm was used. Cubes and tensile specimens were fabricated by LPBF in an argon atmosphere. The laser power was varied from 150 W to 250 W and the scanning speed from 600 to 1200 mm s − 1, while the hatch spacing was kept at 0.1 mm and the layer thickness at 0.03 mm. The build orientation was vertical.

2.2 Post-processing and characterization

Part of the specimens were subjected to hot isostatic pressing at 920 °C for 2 h. Porosity was analysed by X-ray computed tomography (CT). The microstructure was examined using scanning electron microscopy (SEM) and XRD. Tensile tests were performed at room temperature.

Table 1. Process parameters and relative density (DEMO / SYNTHETIC values).

Laser power (W)	Scanning speed (mm/s)	Relative density (%)
150	800	98.9
200	800	99.6
250	800	99.7

3. Results and discussion

Increasing the laser power from 150 W to 250 W reduced the porosity from 1.1 % to 0.3 %. A higher scanning speed increased lack of fusion defects due to insufficient energy input. The as-built microstructure consisted of α' martensite within columnar prior β grains. The yield strength of the as-built specimens was 870 ± 15 MPa and the elongation was 8 %. The α' martensite increased the hardness of the as-built material. HIP closed the internal pores and improved the elongation to 14 %. HIP also transformed the α' martensite into a lamellar α + β microstructure. The build orientation did not significantly affect the tensile strength. Keyhole porosity may be promoted by excessive laser power.

4. Conclusions

Laser power strongly influenced porosity in LPBF Ti-6Al-4V. HIP improved ductility at the cost of a small reduction in yield strength.

References

[1] Placeholder reference entry for the DEMO / SYNTHETIC document.